Technovision 2026-27

Technical Club "FLUX"

Official Rulebook: RoboWalk

Team Composition & Eligibility Rules

1. Team Size: A team may consist of a minimum of 2 member and a maximum of 4 members.
2. Institution Rules: All members of a team must be enrolled students of the same l institution.
3. Single Entry: No participant can be a member of more than one team in this competition.

Bot Specifications (No Circular Wheels Allowed)

1. The bot must not use standard circular wheels, rollers.
2. Any mechanism relying on rolling continuous wheel surfaces for primary motion will result in immediate disqualification.
3. Allowed mechanisms: You can use legs, mechanical joints that walk like legs, or crawling and stepping feet to move your robot.

Dimensions & Volume:

1. The bot must fit within a maximum bounding box of 30 cm × 30 cm × 30 cm at the start of the race.
2. A tolerance of up to 5% on each dimension is allowed.

Weight Limit:

1. The total weight of the bot (including power supply/batteries) must not exceed 3.0 kg.
2. The potential difference (voltage) between any two points on the robot must not exceed 12 V DC.

Control Mechanism:

1. The bot may be **wired** or **wireless** (RF, Bluetooth, or autonomous).
2. For wired bots, Keep the wire loose at all times. Pulling or dragging the robot with the wire will result in a penalty.

Track Specifications & Race Setup

1. The race track consists of a designated lane with a Start Line, Checkpoints, and a Finish Line.

2. The terrain may feature varying surfaces (e.g., smooth acrylic/wood, mild rubber mats, low-profile speed bumps, or slight inclines).
3. Finishing the track from Start to Finish is mandatory to qualify for ranking.

Competition & Track Rules

1. **Restart & Checkpoints:**
    - If your robot flips, gets stuck, or breaks down, you can restart from the last cleared checkpoint.
    - A manual intervention to reset the bot will incur a time penalty.
2. **Technical Timeout:**
    - Each team is allowed **one technical timeout** of up to **2 minutes** for quick repairs or battery replacements during their run.
    - Taking a technical timeout incurs a **10-second penalty** added to the final time.
    - After the timeout, the run resumes from the last cleared checkpoint.

Marking Scheme & Scoring

Winners will be determined based on the Highest Final Score (or Lowest Effective Time).

A. Final Score Formula

$$\text{Total Score} = 600 - \text{TU} - (\text{MNP} \times 5) - (\text{MJP} \times 20)$$

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Where:

- TU**(Time Used):** Total elapsed time taken to clear the track in seconds.
- MNP**(Minor Penalties):** Count of minor infractions (e.g., touching track walls, line step-out, wire pulling).
- MJP**(Major Penalties):** Count of major infractions (e.g., manual resets, technical timeouts, track resets).

Venue Rules & Conduct

1. **Decorum:** All participants are expected to maintain professionalism and sportsmanship at all times within the venue.
2. **Disqualification Grounds:**

- Any violent behavior, ungentlemanly conduct, or abusive language toward judges, volunteers, or opponent teams will lead to **immediate disqualification and removal from the venue**.
- Intentional damage to the competition track, timing equipment, or venue property will result in disqualification and potential financial penalties.

3. **Judges' Authority:** The decision of the track judges and event coordinators is final and binding in all circumstances.